

MCA- Semester I
Flash Test – CS 404: Discrete Mathematics

Date: 25/08/2025

Time: 9:30 – 10:00am

ANSWER ALL OF THE FOLLOWING

1. Write the names of concepts (in one word) the following denote;

- a. (a, b, c, d, e) –
- b. $\{a, b, c, d, e\}$ – Set
- c. $[a, b, c, d, e]$ –
- d. $\exists, \forall$ –

2. Put down one common and one uncommon properties of the following

	Common .	Uncommon
Partition		
Cover		

3. Give the full form of the following acronyms

- a. wff -
- b. CNF –
- c. CSG
- d. 1-1 –
- e. DAG
- f. Poset –

4. Give an example of the following

- a. Recurrence relation
- b. Recursive function
- c. Homomorphism

5. Differentiate between the following

- a. **Characteristic relation and Characteristic function**

CS-404: Discrete Mathematics

Minor-I Test

Date: 25/09/2025

Max Marks: 15

Time: 10:30 – 11:30am

1. List any two utilities of the cardinality of a set for implementation of Set in a computer programme.
2. Given a set say $A = \{a, e, i, o, u, x, y, z\}$ and $r = 3$. What is the meaning of nPr ?
3. For set complementation as an operation, what is the essential requirement for its implementation in a computer programme?
4. Give the formula and the number of characteristic vectors of a set of all digits if first bit is fixed for digit 3 and last bit for digit 6.
5. If 11 integers are randomly selected from the set $\{1, 2, \dots, 20\}$ then prove that there will be at least a pair a, b such that $a - b = 2$.
6. Give the difference between– Set, Sequence and List and one commonality if any.
7. Give the Predicates required to rewrite the following property as a well-formed formula.
8. Write the instruction/expression which is iterated in a method to calculate GCD of two integers a and b . Why is the method called Euclidean Algorithm?
9. Write the statement of strong form of Mathematical Induction.
10. Consider the following statements – For all books if book is cheap then the book is valuable and for all books if book is costly then the book is valuable. It can be inferred that all books are valuable.
 - Represent these statements as wff.
 - Which rule of inference can be used to prove the above?

CS-404: Discrete Mathematics

Minor-II Test

Date: 29/11/2025

Max Marks: 15

Time: 10:30 – 11:30am

Answer any THREE of the following. Each question carries 5 marks

Q1. Let $f: A \rightarrow B$. Give the conditions when f is said to be **defined everywhere** and f is said to be **onto**. Given a set X define and describe identity function on X . Consider an example of a software **E2M** similar to Google translate to translate text from English to Marathi and vice-versa. Give any four Discrete Mathematics concepts that are used in this example.

Q2. Consider a graph $G = (V, E)$. What are V and E ? Give conditions when G is a **graph**, a **regular graph**, and a **complete** graph. Draw the floor-sketch of a house with two rooms, one kitchen, one toilet-cum-bathroom and one storeroom including its doors. Ignore the windows in the sketch. Represent the floor-sketch as a graph. Determine if your floor-sketch includes an Euler's Path or an Euler's circuit or both.

Q3. Consider the given set of transition functions defined as $\delta: Q \rightarrow Q$, $\{q_0, 0 = q_0, \delta q_0, 1 = q_1, \delta q_1, 0 = q_1, \delta q_1, 1 = q_1\}$. Give the 5-tuple describing a finite state machine M . Let L be the set of strings accepted by M . Is $(L, \circ)$ a Semigroup or a Monoid where $\circ$ is string concatenation? Justify your answer.

Q4. Draw the undirected graph $G = (V_1, E_1)$. Let $V_1 = \{a, b, c, d, e, f, g, h, i\}$ and $E_1 = \{(a, b), (a, d), (b, c), (c, i), (c, f), (i, f), (f, e), (d, e), (a, g), (g, h), (g, d)\}$. Give the Hamiltonian path and Hamiltonian circuit if these exist. Given a Hamiltonian graph with k vertices. What is the length of the Hamiltonian path?

School of Computer & Systems Science
CS-404: Discrete Mathematics
Mid-Semester Exam

Date: 10/11/2025
Time: 9:30 – 11:00am

Max Marks: 25

ANSWER QUESTIONS OF **MAXIMUM MARKS 25**.

- 1) Give the set of logical constants. Give one each and describe the well-formed formula for propositional logic and predicate calculus. [3]
- 2) Let C be a subset of a universal set $U = \{u_1, u_2, u_3, \dots, u_n\}$. Name, describe and state application of the function f_A defined as [2]

$$f_A(u_i) = \begin{cases} 1 & \text{if } u_i \in A \\ 0 & \text{if } u_i \notin A \end{cases}$$

- 3) In a given counting problem, how can you decide the use of permutation or combination? A *Hackathon2026* must have **3 PhDs** and **5 MTech** students. Given **24 PhD** and **20 M.Tech.** eligible applicants, how many *Hackathon2026* teams are possible? [3]
- 4) Let $A = \{a, b, c, d\}$, $B = \{1, 2, 3\}$ and $R_1 = \{(a, 2), (b, 1), (c, 2), (d, 1)\}$. Is R_1 a Function? Is R_1^{-1} a function? Justify your answer. [2]
- 5) Given a Boolean alphabet $B = \{0, 1\}$, define a *regular expression* considering the set of standard operators – $\{\circ, +, *\}$. What is the length of the shortest string represented by the $(1 \circ 0)^*(1 \circ 0 + 1^*)(0^* \circ 1)$, justifying your answer. [3+2]
- 6) Differentiate *Relation* from *Function*. Let $A = \mathbb{R}$ and let $F = \{(a, 2a+1) \mid \forall a \in A\}$. What are the properties of F ? Is F a relation or a function? Justify your answer. [5]
- 7) Let $D = \{1, 2, 3, 4, 5, 6\}$ and for given cycles $p_1 = (5, 1, 4)$ and $p_2 = (2, 6, 3)$ complete the following and name the type of function these are.
Give $p_1 \circ p_2$.

a) $(5, 1, 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ - & - & - & - & - & - \end{pmatrix}$

b) $(2, 6, 3) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ - & - & - & - & - & - \end{pmatrix}$ [5]

8) Consider the following and name the basic operators used to compute the set S and matrix $M_1 \odot M_2$. Give the total number of times the basic operators will be executed to compute S and $M_1 \odot M_2$.

i) Let $U = \{a, b, c, d, e, f, g, h, i, j\}$, $G = \{a, b, d, f\}$, $H = \{a, b, c, h, j\}$, $I = \{g, h\}$. To compute, $S = (\overline{G} \cap H) \cup I$.

ii) Let $M_1 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$. Let $\odot$ be the operator

Boolean product $A \odot B$ for $m \times n$ Boolean matrices say A , $m \times p$ matrix and B , $p \times n$ matrix, be denoted by $C = [c_{ij}]$ and defined by,

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 \text{ and } b_{kj} = 1 \text{ for some } k, \\ 0 & \text{otherwise} \end{cases} \quad 1 \leq k \leq p$$

[3+5]

9) Let $R \subseteq Z \times Z$ defined by $(a, b) R (c, d)$ if and only if $a+d = b+c$. Give the properties of R . Give three applications of directed graphs in Computer Science. [2+3]

10) Differentiate the **transitive closure** of an arbitrary set S from an arbitrary relation R . Give the transitive closure of $R_2 = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 2), (3, 4), (4, 1)\}$. [3]

11) Give full account of a **Hasse diagram**. Is *lattice* related to *Hasse diagram*? Justify your answer. [5]

School of Computer & Systems Science
CS-404: Discrete Mathematics
End Semester Examination

Date: 16/12/2025

Max Marks: 50

Time: 10:00 – 12:30am

Both **Part-I [20]** and **Part-II [30]** are compulsory.

Part-I [20]

Answer any **FOUR** of the following. Each carry **5** marks.

1. Consider the two sets $A = \{x \mid x = 4k + 1 \text{ and } k \in \mathbb{N}\}$ and $B = \{x \mid x = 3k + 5 \text{ and } k \in \mathbb{N}\}$.
 - a. Give the value of k such that $|A \cap B| \neq \phi$.
 - b. Let $A \oplus B = (A \cup B) - (A \cap B)$. Draw the Venn diagram and give the cardinality of $A \oplus B$ for $k = 9$.
 - c. Give Venn diagram for $\cup$ **distributive over** $\cap$ for any ^{three} two arbitrary sets X, Y and Z .
2. Let $f: A \rightarrow B$. State the conditions when f is a injective function and when it is a surjective function. Let $\mathbb{N}_n = \{0, 1, 2, \dots, n-1\}$. Consider the $f: \mathbb{N}_8 \rightarrow \mathbb{N}_8$ defined by $f(x) = 2x \bmod 8$. Check if f is **invertible**.
3. Let the relation R_{int} be defined on subset of integers, say S , such that $x R_{int} y$ if and only if $x - y$ is even. Verify if R_{int} is an equivalent relation. Let $[e]$ denote the equivalence class of $e \in S$. If $e, o \in S$ then prove that **either** $[e] = [o]$ **or** $[e] \cap [o] = \phi$.
4. Determine which one of the following is **NOT a well-formed formula**, is **satisfiable** and is a **tautology**,
 - a. $(A \rightarrow B) \rightarrow (C \vee A \rightarrow C \vee B)$
 - b. $(A \rightarrow B) \wedge (\neg B \rightarrow C) \rightarrow (\neg \vee B \rightarrow A)$
 - c. $(A \rightarrow B) \wedge (B \rightarrow C) \rightarrow (C \rightarrow A)$
5. Consider the alphabet $\Sigma = \{1, 0\}$. Give a full description of Σ^* . Let $L \subset \Sigma^*$. Prove that the algebra $(L, \cdot)$ is a Monoid where $\cdot$ is string concatenation?

Part-II [30]

Answer any **THREE** of the following. Each carry **10** marks.

6. a. Define an **(m, n) encoding function**. Is there a relation between m and n ? As per the typical notations what is the meaning of **$e(b)$** . [4]

- b. Let x and y be two bit-strings of length n . Let x be the bit-string based on the encoding function and y be the bit-string received through communication. The errors if any, can be detected by the **Weight** of x , and **Hamming distance** between, say $\delta(x, y)$. Define **Weight** and **Hamming distance**. Give a full account of use of **weight and Hamming distance** for error detection. [6]
7. a. How many distinguishable permutations of the letters in the word **BANANA**, justify your answer. Pigeon-hole Principle. [3]
 b. Let 25 Handbooks of AI in JNU Library contain a total of a total of 62,547 pages. Show that one of the dictionaries must have at least 2,502 pages. [2]
 c. Let $A_1, A_2, A_3, \dots, A_n$ be any n sets. Given the following equation, what is the predicate **P(n)** and values of n such that **P(n)** can be proved as true,

$$\overline{(\cup_{i=1}^n A_i)} = \cap_{i=1}^n \bar{A}_i$$

 Prove P(n) is true. [5]
8. a. Consider the following Table-1 indicating the airline's services and $fare_{ij}$ between the cities $A = \{C1, C2, C3, C4, C5\}$ as specified. Let a relation **R** be defined on A as $(Ci, Cj) \in R$ iff $fare_{ij} \leq 180$. Find **R and verify its properties**. [4]
- Table -1.
- | To
From | C1 | C2 | C3 | C4 | C5 |
|------------|-----|-----|-----|-----|-----|
| C1 | - | 140 | 100 | 150 | 200 |
| C2 | 190 | - | 200 | 160 | 220 |
| C3 | 110 | 180 | - | 190 | 250 |
| C4 | 190 | 200 | 120 | - | 150 |
| C5 | 200 | 100 | 200 | 150 | - |
- b. Represent the relation R above and by two other ways. [3]
 c. Differentiate **Total Ordering** from **Partial Ordering** in an algebra. Give the mathematical structures to represent each. [3]
9. a. Consider a graph $G = (V, E)$. Give conditions when G is a **graph**. Give the features of a **regular graph**, and a **complete** graph. [2.5]
 b. Define the function used to represent sets in a computer program and name the basic operators for computation of set operators. Give the total number of times the basic operators will be executed to compute S, where $U = \{a, b, c, d, e, f, g, h, i, j\}$ the universal set, $G = \{a, b, d, f\}$, $H = \{a, b, c, h, j\}$, $I = \{g, h\}$. To compute, $S = \bar{G} \cap H \cup I$. [6]
 c. Given an arbitrary set A, what is $\wp_A$? Are $|A|$ and $|\wp_A|$ related? Justify your answer. [1.5]

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